
BERT
ALLA RISCOSSA!
Come un modello che sta su un floppy ha imparato a giocare a Doom

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👨‍💻 Open-Source AI orchestration - Haystack @deepset

💙 Small Language Models, Fine-Tuning, RL

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Introducing System One Models & Jev

```
{
  "state": "I've been trying to connect my Stripe
account for 3 days and the integration keeps failing",
  "model": "jev-latest",
  "questions": {
    "department": {
      "type": "choice",
      "instructions": "Which team should handle this",
      "criteria": {
        "billing": "Payment or subscription issues",
        "technical": "Bugs or integration problems",
        "sales": "Pricing or account questions"
      }
    },
  }
}
```

```
{
  "model": "jev-1.13.0",
  "answers": {
    "department": {
      "type": "choice",
      "choice": "technical",
      "confidence": 0.78,
      "probabilities": {
        "technical": 0.85,
        "sales": 0.0,
        "billing": 0.15
      }
    }
  }, ...
}
```

Not every language task needs a generative LLM

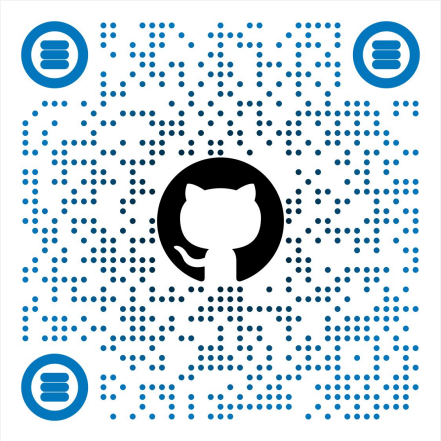
There’s still room for different kinds of language models:

- non-generative
- focused on narrower problems
- faster
- cheaper

And many of them are already available in the open!

Agenda

Materials 👇

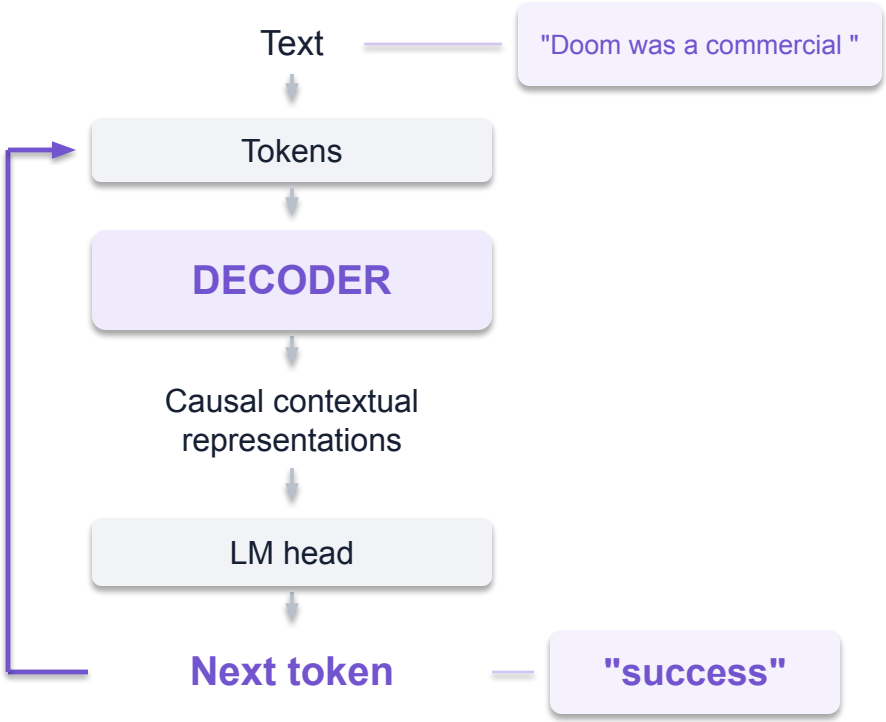


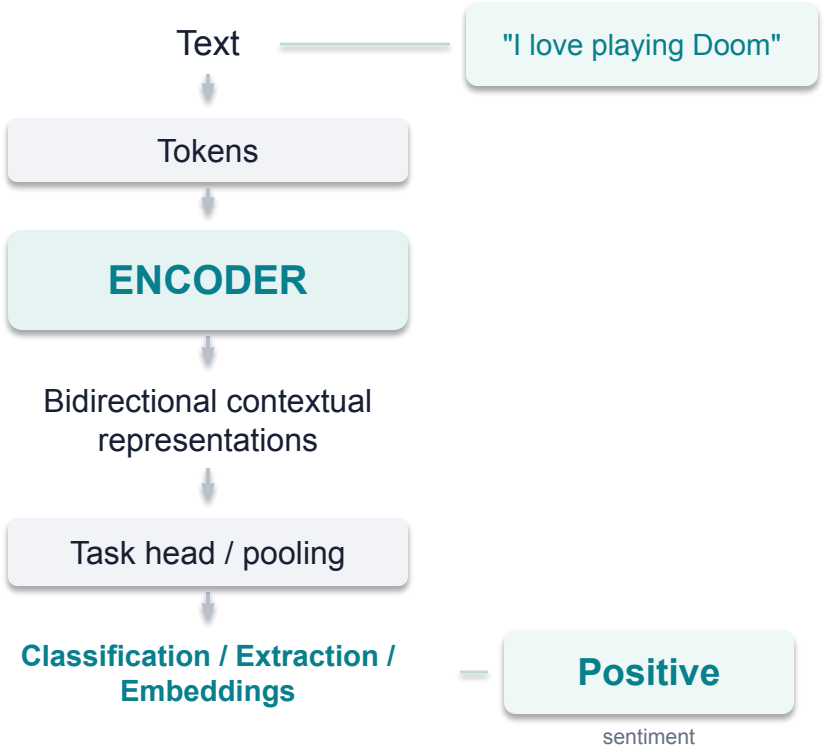
👊 Small encoders and their use cases

🔫 Training a 1M ModernBERT to play Doom

👈 Small encoders and their use cases

Decoders vs Encoders

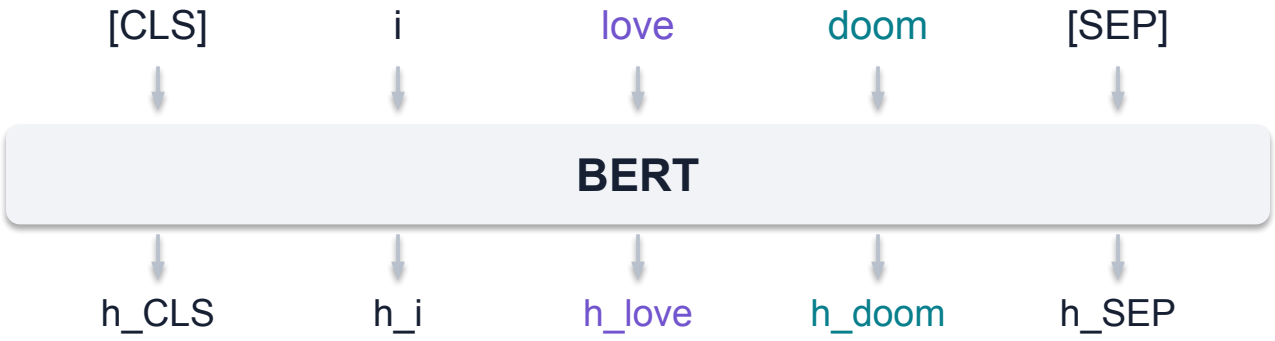




The OG encoder: BERT



- Google, 2018
- Pretrained with Masked Language Modeling + Next Sentence Prediction
- State of the art on many NLP tasks



h_love = [0.784, 0.692, 0.847, …]

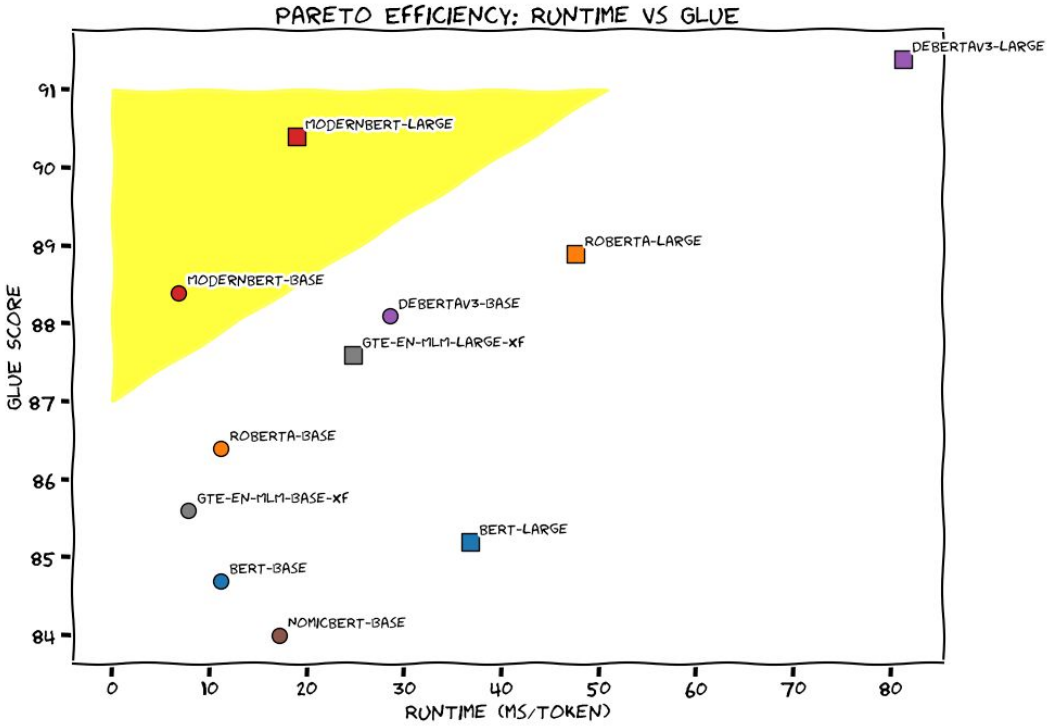
h_doom = [-0.325, -0.234, -0.091, …]

ModernBERT

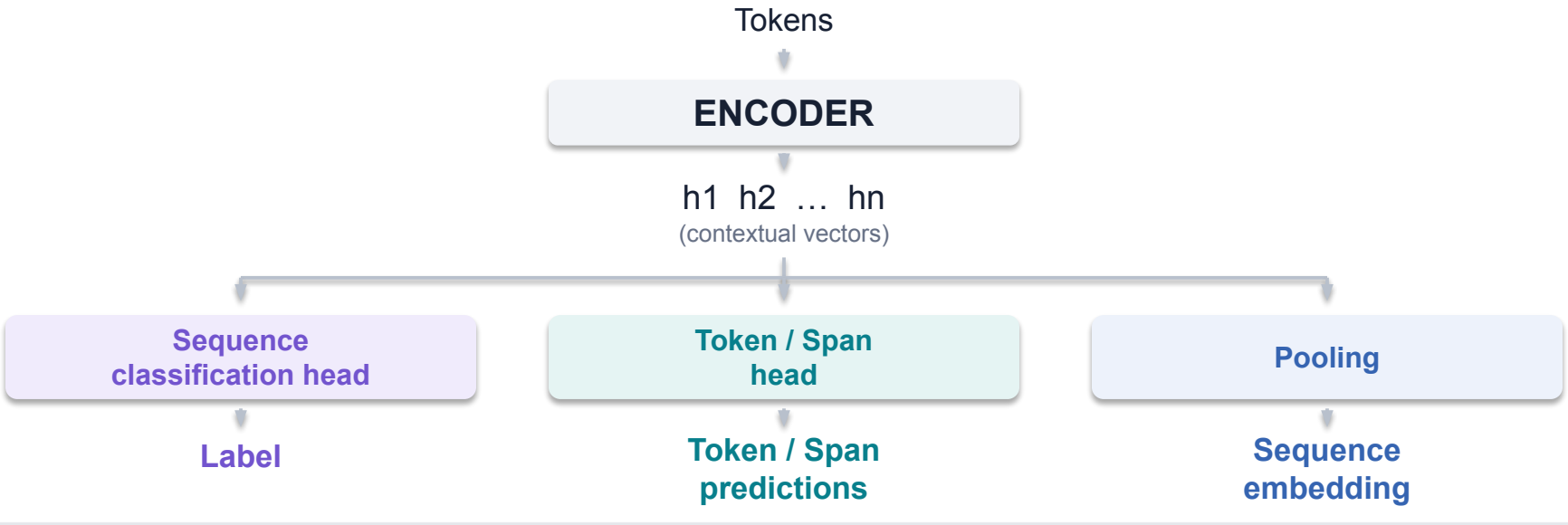
Answer.AI + LightOn, 2024

Advances from LLMs applied to BERT:

- **8k context** instead of 512
- RoPE + GeGLU
- Local / global attention
- FlashAttention + unpadding
- Modern data, including code



Fine-Tuning encoders for different tasks



TYPICAL USE CASES

Sequence classification head	Token / Span head	Pooling
Sentiment classification	Information extraction	Retrieval
Prompt injection detection	NER / PII	
Decisions	Extractive QA	

Sequence classification: sentiment classification

```
from transformers import pipeline

classifier = pipeline(
    "text-classification",
  model="distilbert/distilbert-base-uncased-finetuned-sst-2-english",
)
print(classifier("Doom is an awesome video game."))


# [{'label': 'POSITIVE', 'score': 0.9998100399971008}]
```

Sequence classification: prompt injection detection



Token classification: Named Entity Recognition



Token classification: Personally Identifiable Information Detection



Beyond fixed labels: GLiNER

```python
from gliner2 import GLiNER2

extractor = GLiNER2.from_pretrained("fastino/gliner2-base-v1")

text = """I was out of ammo, so I fought the Imp using the chainsaw.
Suddenly, a demon appeared."""

labels = {
    "weapon": "A weapon used by the player.",
    "enemy": "A monster fought by the player.",
}
print(extractor.extract_entities(text, labels))

# {'entities': {'weapon': ['chainsaw'], 'enemy': ['Imp', 'demon']}}
```

Zero-shot classification with GLiNER

```python
from gliner2 import GLiNER2

extractor = GLiNER2.from_pretrained("fastino/gliner2-base-v1")

text = "Doom crashes every time I load my save. Can someone help me fix it?"

schema = extractor.create_schema().classification(
    "intent",
    {
        "technical_support": "Crashes, errors, or problems running the game.",
        "gameplay_help": "Advice on enemies, weapons, or completing levels.",
        "refund_request": "Requests to return the game and get money back.",
    },
)
print(extractor.extract(text, schema, include_confidence=True))

# {'intent': {'label': 'technical_support', 'confidence': 1.0}}
```

Finding the right fit

SPECIALIZED → **FLEXIBLE**

Task-specific Encoder	GLiNER-like Encoder	Generative LLM
Fixed labels	Dynamic labels	Open-ended tasks
Single forward pass	Single forward pass	Autoregressive decoding
<1B params	<1B params	1B–100B+ params

SMALL / FAST | **LARGER / SLOWER**

Need a decision? Try going small



🔫 Training a 1M ModernBERT to play Doom



The inspiration

- 1.3M ModernBERT encoder
- Trained on 31K human gameplay examples
- Plays Doom *Defend the Center* in real-time on CPU
- My evaluation: 20.38 kills (std dev 5.35)

How SauerkrautLM works

- RGB frame → 40×25 ASCII, character-tokenized
- Depth buffer → 16-bin learned depth embeddings

- 5-layer ModernBERT encoder with two-stage hash embeddings

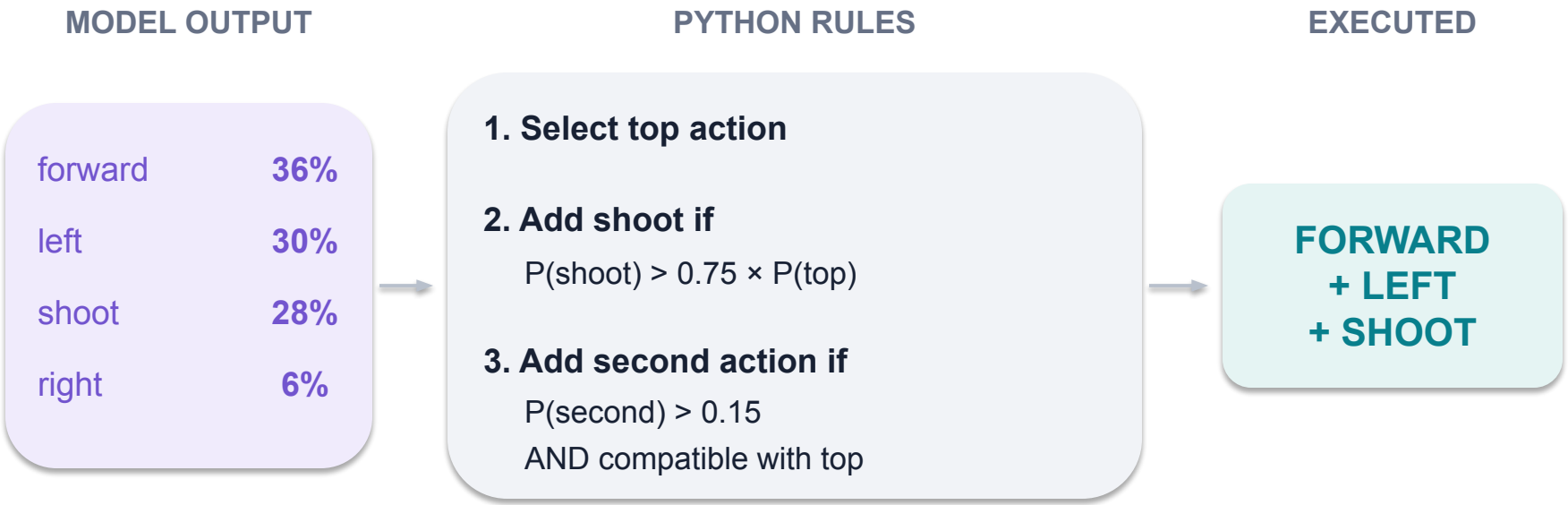
- Attention pooling

- Classification head → probabilities over 4 actions

- Python function combines compatible actions

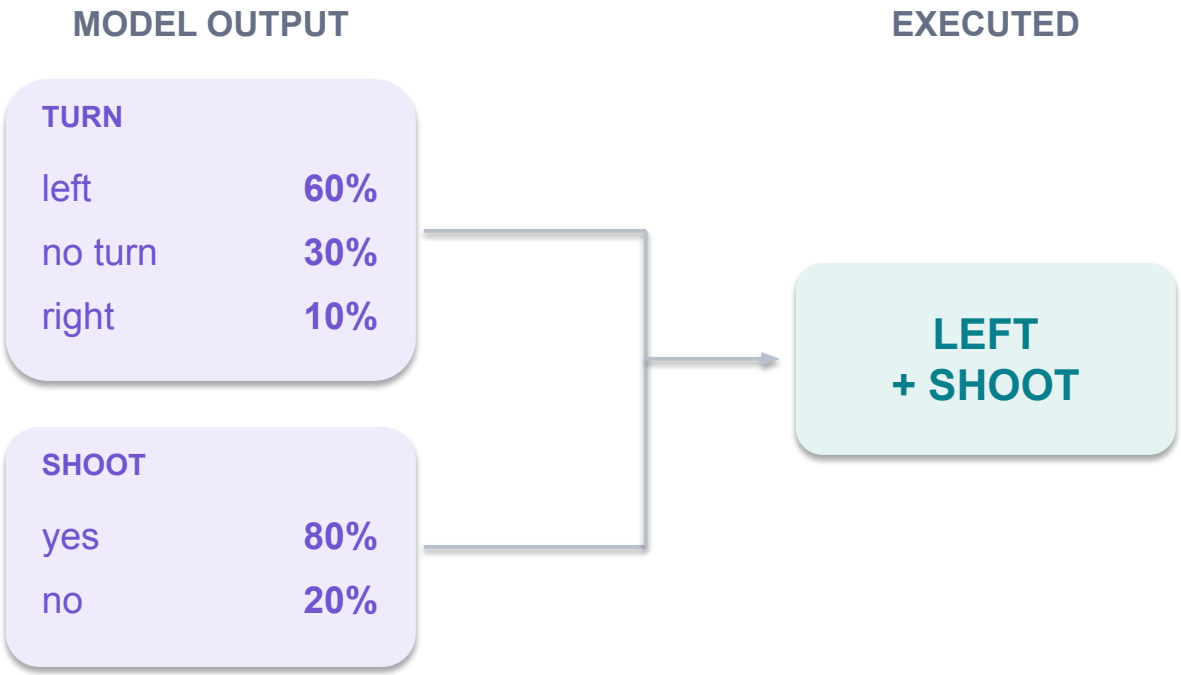


SauerkrautLM - Action prediction



We execute a combination, but we don’t have its probability

Tiny Doom Defender - Action prediction





SauerkrautLM - Input Handling

RGB frame

↓

Grayscale

↓

Resize to 40 × 25

↓

Brightness-to-ASCII mapping

↓

Character tokenization

↓

Token embeddings

Lossy, hand-crafted

VizDoom depth buffer

↓

Resize to 40 × 25

↓

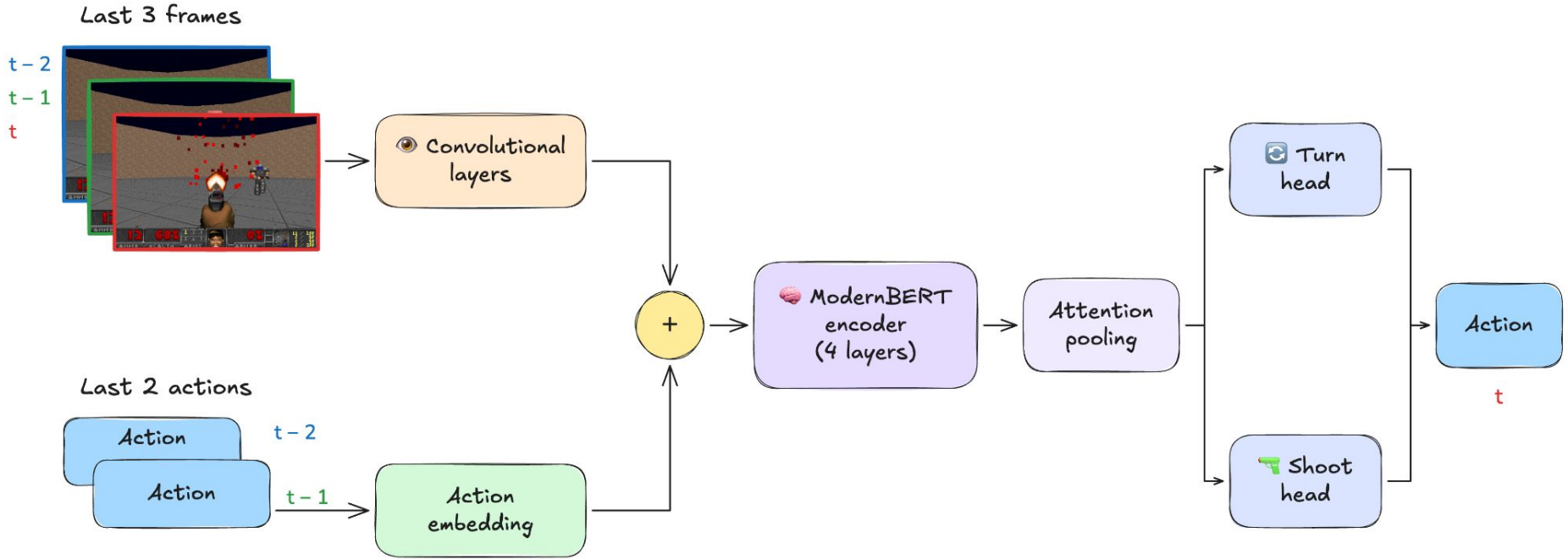
Quantize into 16 bins

↓

Depth embeddings

Privileged info

Tiny Doom Defender - Architecture



How to collect training data?

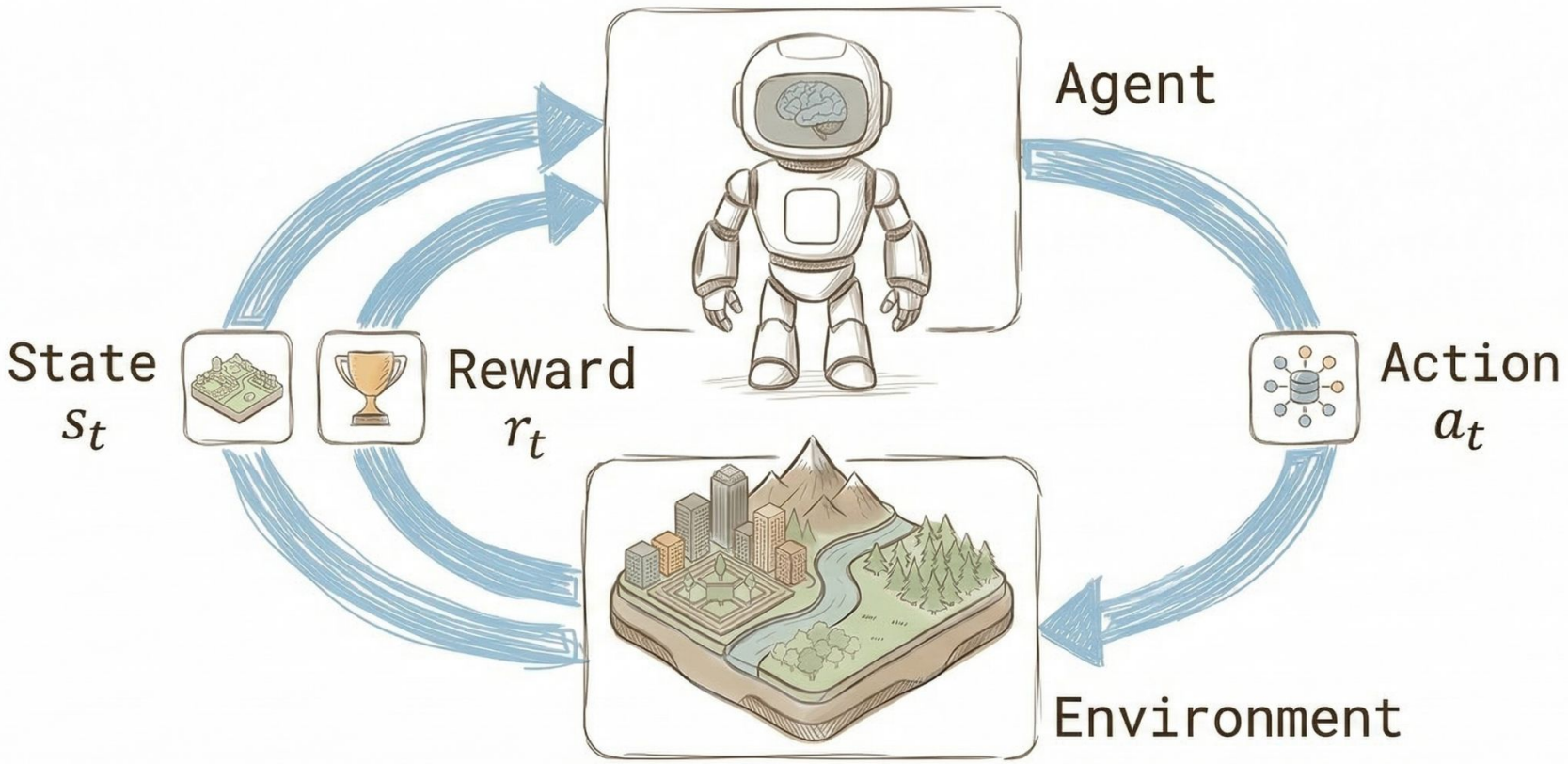


Supervised Fine-Tuning

- Data: 100K examples from the oracle; 90/10 train/eval split
- Loss: sum of shoot and turn cross-entropy losses
- Early stopping: 6 epochs without improvement in joint accuracy
- < 1h on NVIDIA L40 48GB GPU

Model	**Mean kills**	**Std dev**
SauerkrautLM-Doom-MultiVec-1.3M	20.38	5.35
tiny-doom-defender-sft	22.19	3.76
Oracle*	23.05	2.85

Reinforcement Learning



PPO training

- Start from the SFT Model
- Train the last encoder block, attention pooling and heads

- Small hyperparameter search
  - **Train**: 30 iterations
  - **Validation**: evaluate all checkpoints; select the best
  - **Test**: evaluate the best checkpoint on 1000 episodes

  - Separate seeds for train/validation/test

- Each round: < 2h on MacBook M4 PRO



Results

Model	Mean kills	Std dev
SauerkrautLM-Doom-MultiVec-1.3M	20.38	5.35
tiny-doom-defender-sft	22.19	3.76
Oracle*	23.05	2.85
tiny-doom-defender	**23.12**	**2.81**

Tiny Doom Defender is smaller, simpler, faster and killer!

With int8 quantization, it fits on a floppy disk 💾

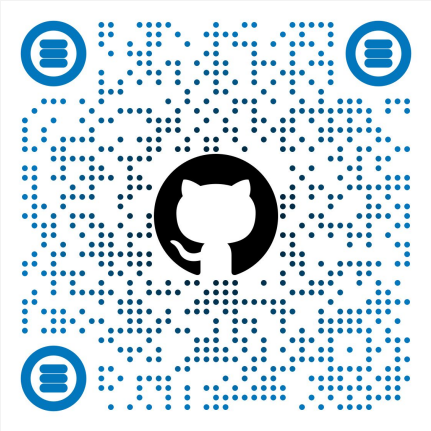
🕹️ Demo time



👾 BERT alla riscossa - Come un modello che sta su un floppy ha imparato a giocare a Doom

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Talk Materials



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